SDI-12 Sensor Station - System Workflow

Board: Arduino Zero / MKR / SAMD21 (TC4 hardware timer). This document explains how main.ino and all modules work together.

1. Big picture (two separate sensor paths)

- REAL-TIME path: instant I2C reads for TFT, SDI-12, and datalog.csv
- AVERAGED path: TC4 timer every 2 s -> avg_datalog.csv only

```
+------------------+
                    |   main.ino       |
                    |  setup() / loop()|
                    +--------+---------+
                             |
         +-------------------+-------------------+
         |                                       |
         v                                       v
   REAL-TIME (on demand)                   AVERAGED (timer-driven)
   - Dashboard: readSensors()              - SensorSampler (TC4 ISR)
   - SDI-12: readSensors() on M/D1/D2/R0   - AvgDataLogger -> avg_datalog.csv
   - DataLogger: readSensors()             - Every 2 seconds
     -> datalog.csv every 60 s
```

2. Module map (what each file does)

File	Role
main.ino	Entry point. Calls all init and loop handlers in order.
HardwareConfig.h	Pins, intervals: dashboard 1s, datalog 60s, sample 2s.
SensorReading.cpp	BME280 + BH1750 I2C. readSensors(), getSensorData().
SensorSampler.cpp	TC4 ISR (10ms base, 2s sample). Builds 2s averages.
SdCard.cpp	One SD card + RTC shared by both loggers.
DataLogger.cpp	datalog.csv - periodic (60s) + manual button.
AvgDataLogger.cpp	avg_datalog.csv - one row per 2s average.
Dashboard.cpp	TFT display - real-time readSensors(), NOT averages.
sdi12.cpp	SDI-12 slave on Serial1. Uses getSensorData() after read.

3. setup() order (runs once at power-on)

```
Serial.begin(9600)
  sensorsInit()          // Wire + BME280 + BH1750
  sdCardInit()           // RTC + SD card (shared)
  sensorSamplerInit()    // Start TC4 timer (SAMD21)
  dataLoggerInit()       // Buttons + datalog.csv header
  avgDataLoggerInit()    // avg_datalog.csv header
  dashboardInit()        // TFT labels
  sdi12Init('0', pin 8)  // SDI-12 bus
  readSensors()          // First reading
```

4. loop() order (runs forever)

```
1. sensorSamplerService()  // FIRST: process 2s timer flags, I2C for averages
  2. sdi12Handle()           // SDI-12 commands + R0 stream (1s)
  3. avgDataLoggerUpdate()   // Write avg_datalog.csv when new 2s average ready
```

```
    4. dataLoggerUpdate()      // 60s log + manual/clear buttons
    5. dashboardUpdate()       // TFT refresh every 1s, real-time readSensors()
```

5. Averaged path detail (SensorSampler)

- TC4 hardware interrupt fires every ~10 ms (base tick).
- ISR counts 200 ticks -> one sample request every 2000 ms (2 s).
- ISR only sets g_sampleTicksPending (NO I2C inside interrupt).
- sensorSamplerService() in loop reads sensors and accumulates.
- After 1 sample in window -> finalizeAverageWindow().
- AvgDataLogger writes row and calls sensorSamplerAcknowledgeAverage().

TC4

```
        |                              |
        +-- tick divider (x200)        +-- readSensors() if pending
        +-- g_sampleTicksPending++     +-- finalize -> newAverageAvailable
                                               |
                                               v
                                     avgDataLoggerUpdate()
                                               |
                                               v
                                     avg_datalog.csv (every ~2s)
```

6. Real-time path detail (everything else)

- Dashboard: every 1000 ms calls readSensors() then getSensorData().
- DataLogger: every 60000 ms OR button pin 2 -> datalog.csv row.
- SDI-12: host sends command ending with ! -> readSensors() -> reply.
- SDI-12 R0 stream: every 1000 ms while streaming -> readSensors().
- Clear SD button (pin 3): deletes datalog.csv only (not avg file).

7. CSV files (do not mix them up)

File	Interval	Data type
datalog.csv	60 s + manual	Instant snapshot at log time
avg_datalog.csv	~2 s	Hardware-timed average (1 sample/window)

8. SDI-12 commands (address 0 by default)

- ? or 0! -> Respond with sensor address
- 0M! -> Measure: returns count of parameters
- 0D1! -> Send BME: temp, humidity, pressure
- 0D2! -> Send BH1750: lux
- 0R0! -> Send all + start continuous stream every 1 s
- 0An! -> Change address to n

9. Timing constants (HardwareConfig.h)

- kSensorSampleMs = 2000 ms (Average path: time between I2C samples)
- kSensorAverageMs = 2000 ms (Length of one average window)
- kTc4BasePeriodMs = 10 ms (TC4 hardware tick (divider -> 2 s))
- kDashboardRefreshMs = 1000 ms (TFT update rate)
- kLogIntervalMs = 60000 ms (datalog.csv auto-log interval)
- R0_STREAM_INTERVAL_MS = 1000 ms (SDI-12 R0 stream (in sdi12.cpp))

10. Common confusion - quick answers

Q: Why two sensor reads?

A: San

Q: Does TFT show averages?

A: No.

Q: Does SDI-12 use averages?

A: No.

Q: What does TC4 do?

A: Onl

Q: Non-SAMD board?

A: Ser

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